Faculty of Information and Communication Technology
ICT First Years and Foundation Unit

Tshwane University of Technology
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Principles of Programming and Introduction to Programming (Extended) (Year 1)

PPAF05D & TROF05D
CLASS TEST 2 A

30 May 2025 Duration: **90 min** Total: **35** Full Marks: **35** Number of Pages: **8**	Examiner: VL Baloyi Moderator: KS Tshehla

Test Instructions

1. **Read the instructions** at the start of each question carefully.
2. **Answer all questions** on the **Electronic Campus (EC)** platform.
3. Use **Notepad++ and the console only** — **no IDEs** (like Eclipse, IntelliJ, etc).
4. **Test (debug)** your code before submitting your answers.
5. **Copy and paste** your final answers into EC (**do not cut** and paste).
6. **Keep a backup** of each program in Notepad++ until you've confirmed your submission.
7. Only scientific, non-programmable calculators are allowed.
8. **Cellphones are not allowed** during the test.
9. **No sharing** of calculators or stationery.

Question 1 (TRUE or FALSE) [5]

State whether the following statements are true or false.

	Statements
1.1	String is a primitive data type in Java.
1.2	The expression x++ increases the value of x by 1 after it is used in an expression.
1.3	result+=10 is a valid compound assignment statement
1.4	The method print() moves the cursor to the next line after printing.
1.5	Java code must be written inside a method named start().

Question 2 (Multiple choice) [5]

2.1	What is the correct way to declare a constant for tax percentage?
	A. final double TAX = 0.15;
	B. const double tax = 0.15;
	C. TAX = final 0.15;
	D. double final TAX = 0.15;
2.2	What is the correct method to read a String from the user?
	A. readLine()
	B. keyboard.nextLine()
	C. input.nextString()
	D. String();
2.3	Which of the following is a valid identifier?
	A. 1number
	B. total_value
	C. float
	D. @amount
2.4	What will be the result of this operation: 7 / 2 in Java using integers?
	A. 3
	B. 3.5
	C. 4
	D. 0
2.5	What does the Scanner class do?

	A. Prints text to the screen
	B. Compiles the Java code
	C. Stores constants
	D. Reads input from the keyboard

Question 3 (SHORT ANSWERS) **[5]**

One-line answers only. Do not write more than that as it will be marked wrong.

3.1	What is the purpose of the main() method in a Java program?	**(1)**
3.2	What does the .nextInt() method return?	**(1)**
3.3	What is the difference between print() and println()?	**(2)**
3.4	What keyword is used to define a class in Java?	**(1)**

Question 4 (Predict the output) **[5]**

Given a = 9, b = 4, d = 3, show all calculations to determine the value of x, where:

x = a \ b mod 5 * d ^ 2

A	
B	
C	
D	
E	

Question 5 (Write pseudocode) **[5]**

Musa is a student assistant working part-time at the university library. Each month, he records the number of hours he worked and multiplies it by the hourly pay rate to calculate his salary.

You only need to fill in the letters indicated with the correct code.

Write **pseudocode** for a program that(The class name is **SalaryCalculator**):

- Asks Musa to enter his **name**.
- Asks how many **hours** he worked this month.
- Asks for his **pay rate per hour**.
- Calculates his **total salary**.
- Displays a message showing **his name** and the **salary amount** he will receive.

Example:

```
Please enter your name: Koma
How many hours did you work this month? 80
What is your pay rate per hour? 900
Koma, your total salary for this month is: R72000.0
```

```
A
begin
        B
        begin
                //Declare variables
                String sName
                int iHoursWorked
                double rPayRate, rSalary

                //Input
                C
                D
                E
                F
                G
                H

                //Process
                I

                //Output
                J
        end
end
```

Question 6 (Spot and Fix the Error) **[5]**

The following Java program allows the user to enter their product's price, then compares it to a random competitor price between R1 and R5000. It should display the higher price, formatted to two decimal places.

However, the code below contains multiple errors. Your task is to identify and correct them.

```
import java.util.Scanner;

public class PriceBattle
{
   public static void main(String[] args)
   {
      //Declare
      Scanner scan = new Scanner(System.in);
      DecimalFormat df = new DecimalFormat("#.00");
      double userPrice, compPrice, max;

      //Input
      System.out.print("Enter your price: ");
      userPrice = scan.nextdouble;

      //Randomize price
      compPrice = 1 + Math.random(5000);

      //Get the maximum
      max = math.min(userPrice, max);

      System.out.println("Higher price is R " + df.format(Max));
   }
}
```

Question 7 (Write Java code) **[5]**

Thuli wants to create a program that calculates the **average of three test marks** and **displays the result** to the user. She also wants the result to show **two decimal places**.

Write a complete Java program that:

- Asks the user to **enter their name**
- Asks for **three test marks** (**integers**)
- Calculates the **average** of the three marks (the result must be a **double**, use casting if necessary)
- Displays the **name** and the **average**, rounded to **two decimal places**

You may use either:

- DecimalFormat
  or
- Math.round() with manual division to round to two decimal places.